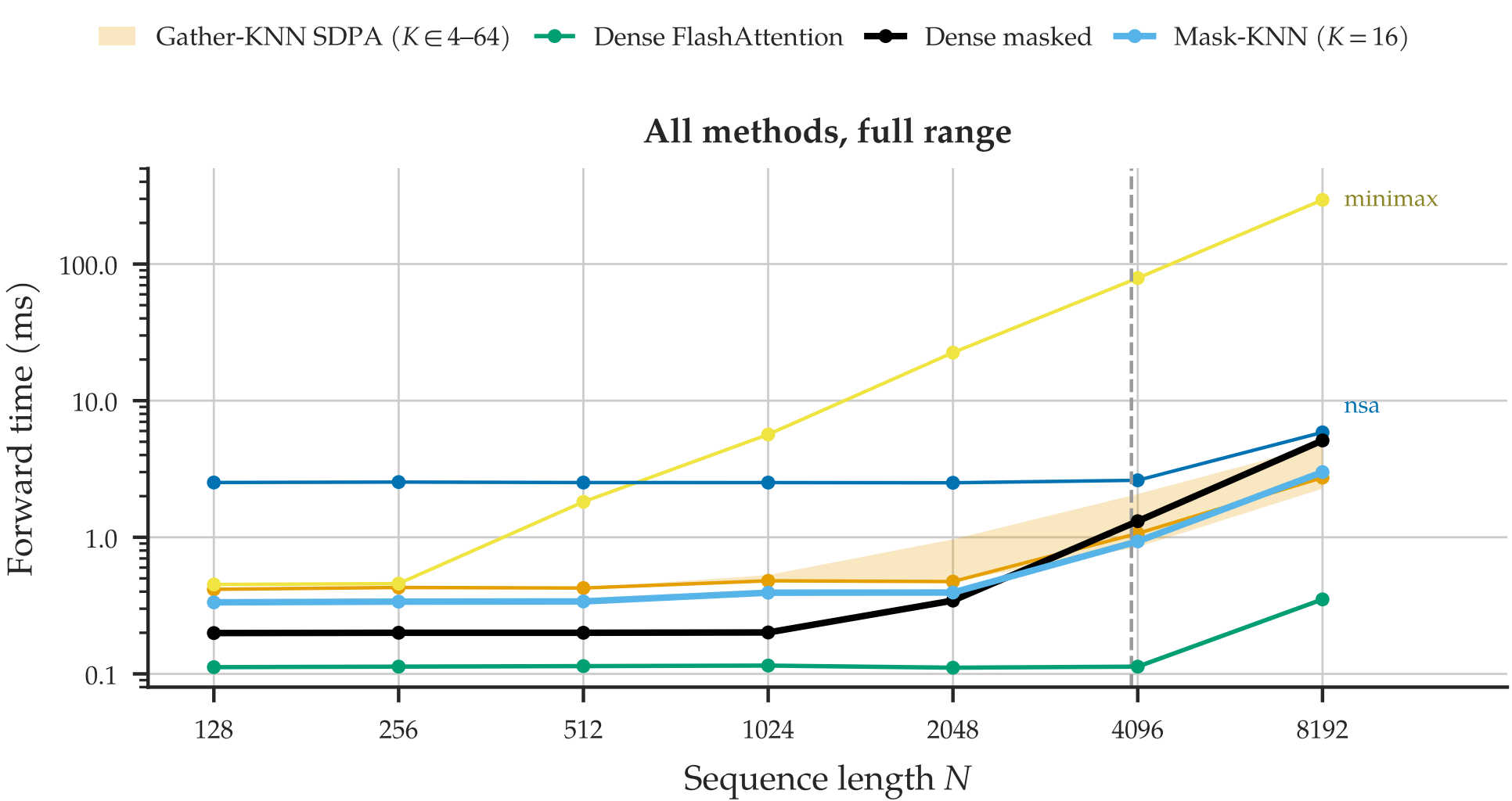
Gather-KNN SDPA ($K \in 4$–64)
Dense FlashAttention
Dense masked
Mask-KNN ($K = 16$)
All methods, full range
minimax
nsa
Forward time (ms)
100.0
10.0
1.0
0.1
128
256
512
1024
2048
4096
8192
Sequence length N